December 7, 2022

APS110 – Lecture 35

See hand written notes for drawings.

Enthalpy of phase chages
→ heat required for a change of phase related to the work required to push back the atmosphere

- Starting at -20°C, as you apply heat this heat will cause a steady change of temperature
- The slope will depend on the material that you're heating and it will depend on the phase that it's in
- At a temp where a phase transformation occurs, you can apply heat, but this heat will be consumed by the phase transformation
- Phase transformation from liquid to vapour is much larger (41kJ per mol of water) than from ice to liquid (6kJ/mol)
    - Enthalpy of vapourization
- What does the slope represent?
    - Determined experimentally
    - The slope of liquid phase is the shallowest and the slope of the gas phase is the steepest
    - Slope is the change in temp for an applied heat (~ °C/J)
        - Thus, we can say that $q = (1/slope) * \Delta T$
        - The (1/slope) has units of J/K mol → MOLAR HEAT CAPACITY

$$\mathbf{q = n\ Cp\ \Delta T}$$
$$\mathbf{q = m\ c\ \Delta T}$$

- Gibb's energy is just a restatement of the second law of thermodynamics (entropy of universe is always increasing)

Phase Equilibria

- Phase diagrams
    - Water is unique, because it has a negative slope in its solid-liquid phase boundary
- Often, we <u>fix the pressure at **1 atm**</u> and consider changes to **temperature**, or consider changes in **composition**
    - Recall slushie demo from last class → water-sugar system (concentration of sugar in water)

- Consider water-sugar system
    - Plot temperature [°C] vs. composition [wt%]
        - We call it 'weight percent' but it's dealing with ***mass***
    - 0 wt% sugar = pure water
    - At warm temperatures (~80°) you can dissolve a LOT of sugar

        - At higher temps you can dissolve more
    - At a certain point though, the sugar will precipitate out (no more will dissolve)
        - You will have liquid syrup and solid sugar
        - This chart will thus have a **phase diagram**
    - You will notice that because of increased sugar concentration you get **freezing point depression**
    - There is a horizontal line at the bottom of the diagram → **isotherm**
        - Solid (ice) + solid (sugar)
    - There is also a **"slush"** component, which is solid (ice) + liquid (syrup)
        - Slush is simply ice (0 wt% sugar) and syrup (high wt% sugar)
        - This is a MASS BALANCE
            - You put in a certain amount of sugar (ex. 20 wt%)
            - The water that solidifies to ice takes none of the sugar with it, so the entirety of the 20 wt% sugar is in the syrup
        - There is a mass fraction, of solid vs. liquid
            - These phase diagrams are just playing around with mass balance
        - As you raise the temperature (for a given conc.), we are approaching the phase boundary, with the completely liquid (syrup) phase
            - As such, it makes sense that there would be a greater fraction of **liquid**
        - As you lower the temperature, the fraction of **solid** increases
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